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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 05-03-2026

Automated Quiz Engine with PDF Certification Generation

SDG: SDG 4 – Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- In the digital education era, online assessments are becoming increasingly important.
- Many institutions still rely on manual quiz evaluation and certificate preparation.
- This consumes time and increases the chances of human errors.
- There is a need for an automated system to conduct quizzes efficiently.
- An automated quiz engine can evaluate answers instantly.
- Generating certificates manually is also time-consuming.
- This project provides a complete automated solution for quizzes and certification.
- It enhances learning experience with fast evaluation and instant PDF certificate generation.

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PROBLEM STATEMENT



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- Many existing quiz systems require manual correction and certificate preparation.
- This leads to delays in result publication and certificate distribution.
- Manual systems are prone to errors and data mismanagement.
- There is no centralized system to manage quiz records and certifications.
- Institutions face difficulty in maintaining performance records.
- Students expect quick results and digital certificates.
- Hence, there is a need for a reliable automated quiz and certification system.
- The system should ensure accuracy, speed, and secure data handling.

OBJECTIVE(S)



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- To design and develop an automated quiz engine with instant evaluation.
- To implement automatic PDF certificate generation for successful candidates.

PROPOSED SYSTEM



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- Users can register and log in securely.
- Admin can create and manage quizzes dynamically.
- Questions are stored and retrieved from the database.
- The system evaluates answers automatically.
- Scores are calculated instantly after submission.
- Qualified users receive auto-generated PDF certificates.
- All records are securely stored in the database.
- The system reduces manual work and improves efficiency.

PROPOSED SYSTEM ARCHITECTURE



Explanation:

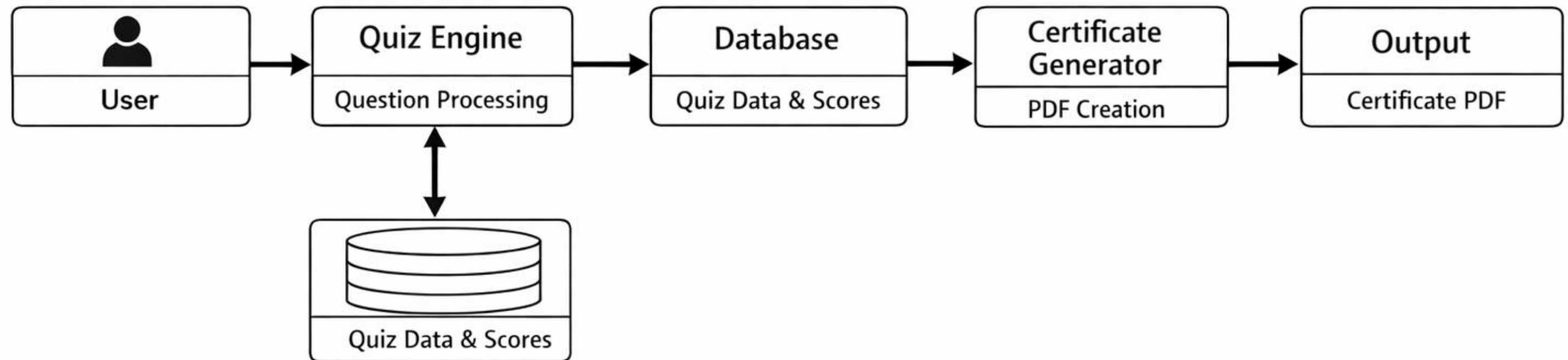
Input: User registration details, quiz answers, admin question entries.

Processing: Authentication, answer validation, score calculation, certificate generation.

Database: Stores user data, quiz questions, scores, and certificate records.

Output: Quiz results, score display, downloadable PDF certificate.

WORKFLOW



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TECHNOLOGIES USED



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Frontend: HTML, CSS, JavaScript

Backend: Java (Spring Boot / Servlets & JSP)

Database: MySQL

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Tools: VS Code

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Hardware Requirements: Minimum 4GB RAM, i3 Processor

Software Requirements: JDK 8+, MySQL Server, Web Browser

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MODULE DESCRIPTION



Module 1: User Management

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Handles user registration, login, authentication, and profile management.

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Module 2: Quiz Management

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Admin creates, updates, and deletes quiz questions and sets difficulty levels.

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Module 3: Evaluation Engine

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Automatically checks answers and calculates scores instantly.

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Module 4: PDF Certification Module

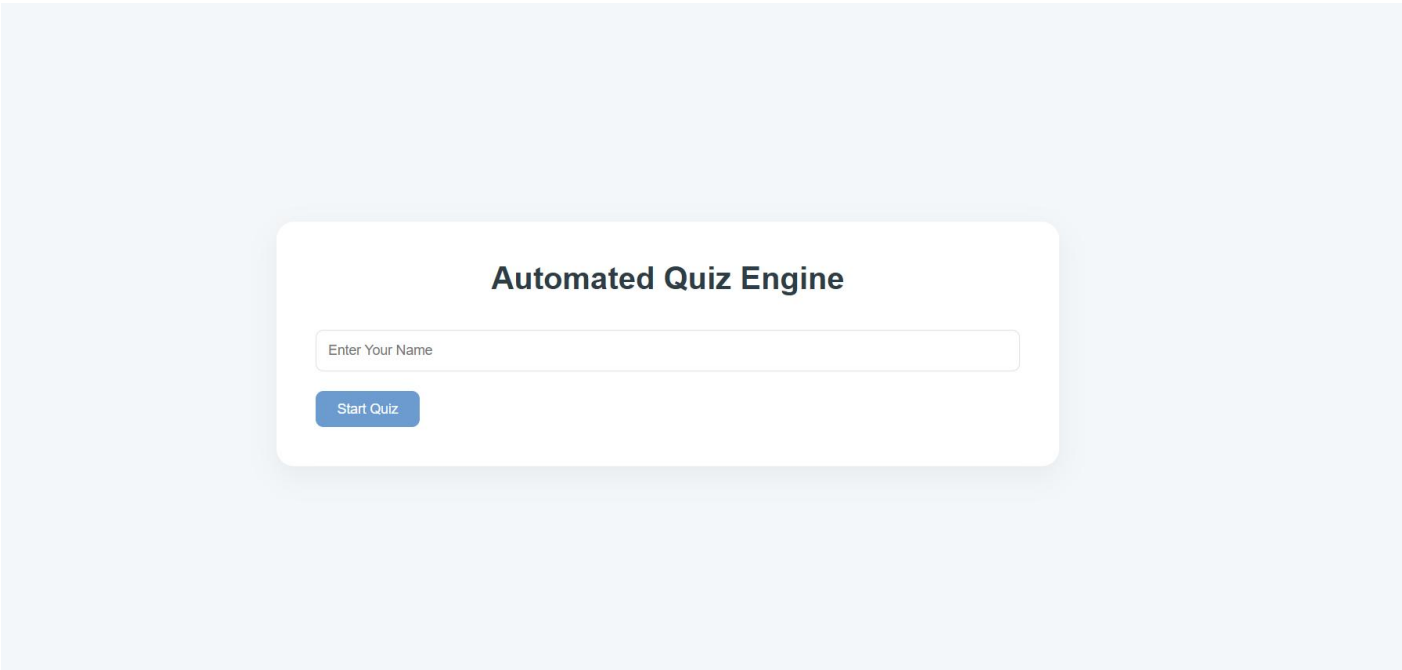
9

Generates personalized PDF certificates with user name, score, and date.

IMPLEMENTATION SCREENSHOTS



Home Page: Displays project title.



Main Quiz Page: Displays multiple-choice questions and timer .



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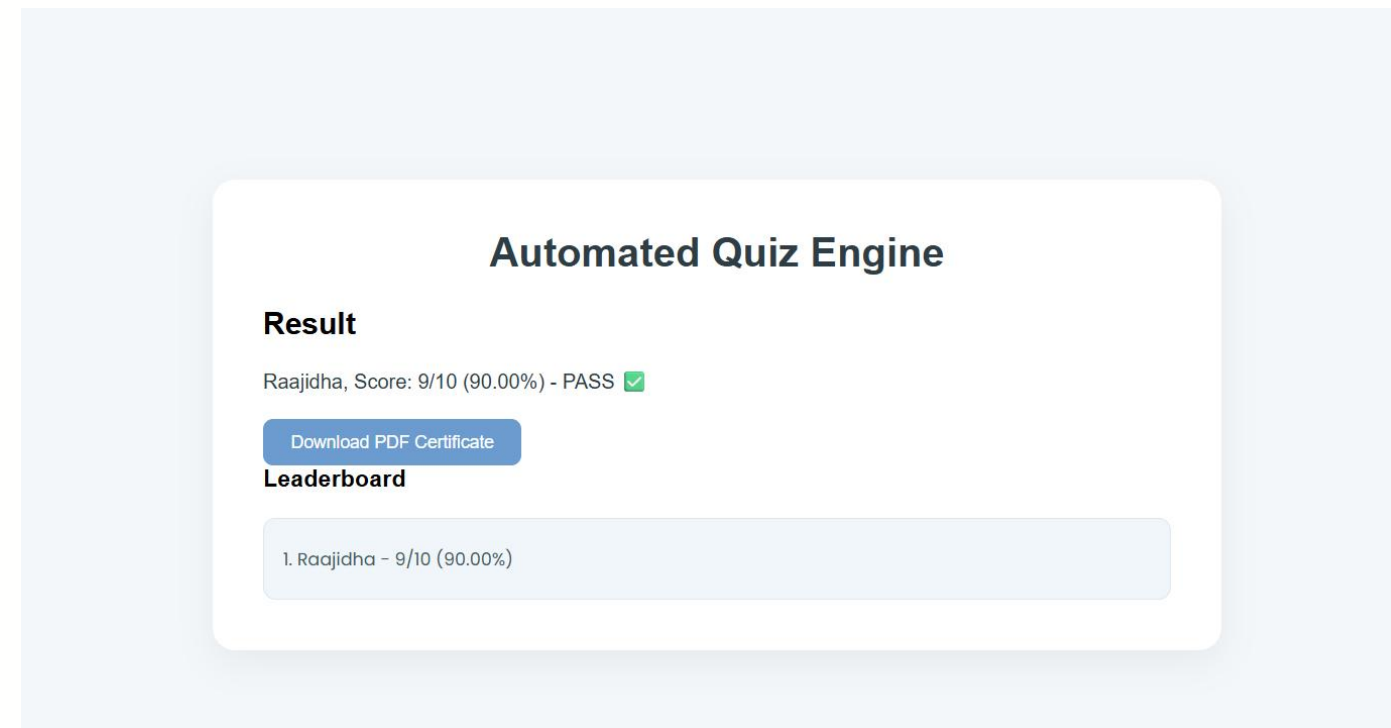
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IMPLEMENTATION SCREENSHOTS



Result Page: Shows total score and qualification status instantly after submission.



Certificate Page: Displays downloadable PDF certificate for qualified users.



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ADVANTAGES & LIMITATIONS



Advantages:

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Reduces manual effort

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Instant evaluation and result generation

3

Automatic PDF certificate creation

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Secure data storage

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User-friendly interface

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Limitations:

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Requires internet connection

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Limited to objective-type questions

Initial setup and maintenance required

Thank you

